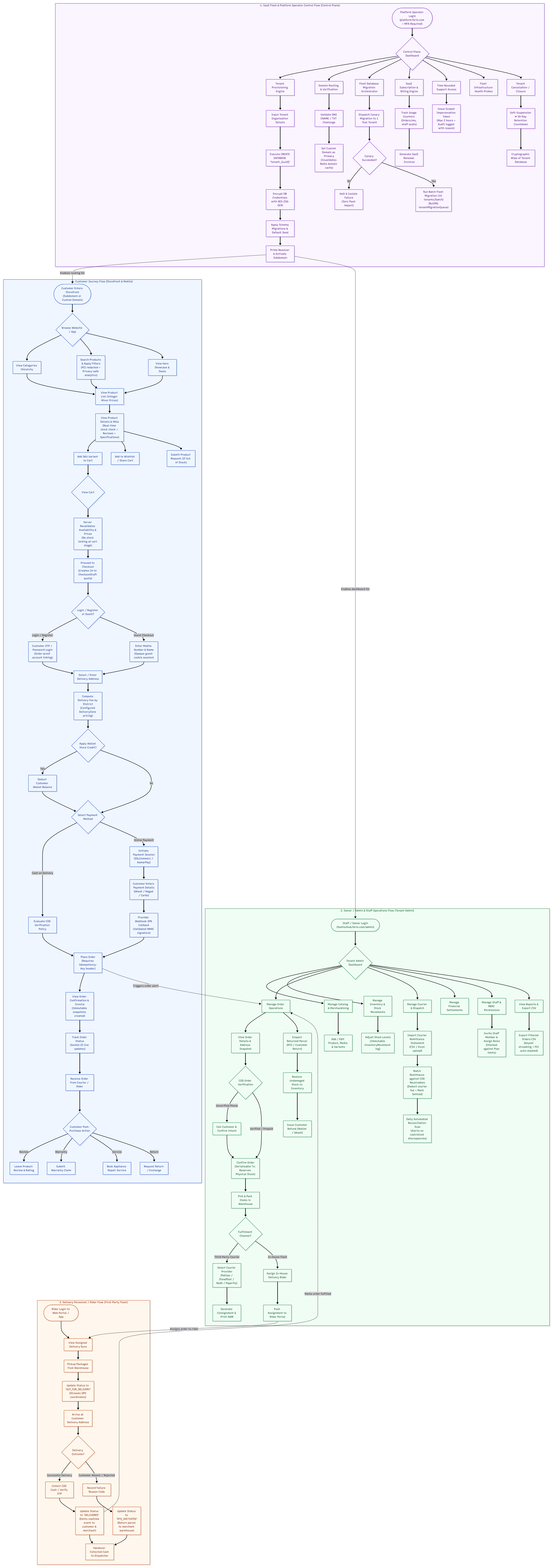
4. SaaS Fleet & Platform Operator Control Flow (Control Plane)
Platform Operator Login (platform.ferio.com ▪ MFA Required)
Control Plane Dashboard
Tenant Provisioning Engine
Input Tenant Organization Details
Execute CREATE DATABASE 'tenant_{uuid}'
Encrypt DB Credentials with AES-256-GCM
Apply Schema Migrations & Default Seed
Prime Resolver & Activate Subdomain
Domain Routing & Verification
Validate DNS CNAME / TXT Challenge
Set Custom Domain as Primary (Invalidates Redis domain cache)
Fleet Database Migration Orchestrator
Dispatch Canary Migration to 1 Test Tenant
Canary Succeeded?
No
Halt & Isolate Failure (Zero fleet impact)
Yes
Run Batch Fleet Migration (10 tenants/batch) (BullMQ tenantMigrationQueue)
SaaS Subscription & Billing Engine
Track Usage Counters (Orders/mo, staff seats)
Generate SaaS Renewal Invoices
Time-Bounded Support Access
Issue Scoped Impersonation Token (Max 2 hours ▪ Audit logged with reason)
Fleet Infrastructure Health Probes
Tenant Cancellation / Closure
Soft-Suspension ➔ 30-Day Retention Countdown
Cryptographic Wipe of Tenant Database
Enables routing for
Enables dashboard for
1. Customer Journey Flow (Storefront & Mobile)
Customer Enters Storefront (Subdomain or Custom Domain)
Browse Website / App
View Categories Hierarchy
Search Products & Apply Filters (PII redacted ▪ Privacy-safe analytics)
View Hero Showcase & Deals
View Product List (Integer Minor Prices)
View Product Details & SKUs (Real-time stock check ▪ Reviews ▪ Specifications)
Add SKU Variant to Cart
Add to Wishlist / Share Cart
Submit Product Request (If Out of Stock)
View Cart
Server Revalidates Availability & Prices (No stock locking at cart stage)
Proceed to Checkout (Creates 24-hr CheckoutDraft quote)
Login / Register or Guest?
Login / Register
Customer OTP / Password Login (Order-proof account linking)
Guest Checkout
Enter Mobile Number & Name (Opaque guest cookie session)
Select / Enter Delivery Address
Compute Delivery Fee by District (Configured DeliveryZone pricing)
Apply Wallet Store Credit?
Yes
Deduct Customer Wallet Balance
No
Select Payment Method
Cash on Delivery
Evaluate COD Verification Policy
Online Payment
Initiate Payment Session (SSLCommerz / AamarPay)
Customer Enters Payment Details (bKash / Nagad / Cards)
Provider Webhook IPN Callback (Validated HMAC signature)
Place Order (Requires Idempotency-Key header)
Triggers order alert
View Order Confirmation & Invoice (Immutable snapshots created)
Track Order Status (Socket.IO live updates)
Receive Order from Courier / Rider
Customer Post-Purchase Action
Review
Leave Product Review & Rating
Warranty
Submit Warranty Claim
Service
Book Appliance Repair Service
Return
Request Return / Exchange
2. Owner / Admin & Staff Operations Flow (Tenant Admin)
Staff / Owner Login (fashionhub.ferio.com/admin)
Tenant Admin Dashboard
Manage Order Operations
View Order Details & Address Snapshot
COD Order Verification
Unverified Phone
Call Customer & Confirm Intent
Verified / Prepaid
Confirm Order (Serializable Tx: Reserves Physical Stock)
Pick & Pack Items in Warehouse
Fulfillment Channel?
Third-Party Courier
Select Courier Provider (Pathao / Steadfast / RedX / Paperfly)
Generate Consignment & Print AWB
In-House Fleet
Assign In-House Delivery Rider
Push Assignment to Rider Portal
Inspect Returned Parcel (RTO / Customer Return)
Restore Undamaged Stock to Inventory
Issue Customer Refund (Wallet / bKash)
Marks order fulfilled
Manage Catalog & Merchandising
Add / Edit Product, Media & Variants
Manage Inventory & Stock Movements
Adjust Stock Levels (Immutable InventoryMovement log)
Manage Courier & Dispatch
Import Courier Remittance Statement (CSV / Excel upload)
Match Remittance against COD Receivables (Deduct courier fee ▪ Mark Settled)
Daily Automated Reconciliation Scan (Alerts on cash/stock discrepancies)
Manage Financial Settlements
Manage Staff & RBAC Permissions
Invite Staff Member & Assign Roles (Checked against Plan limits)
View Reports & Export CSV
Export Filtered Orders CSV (Keyset streaming ▪ PII auto-masked)
Assigns order to rider
3. Delivery Personnel / Rider Flow (First-Party Fleet)
Rider Login to Web Portal / App
View Assigned Delivery Runs
Pickup Packages from Warehouse
Update Status to 'OUT_FOR_DELIVERY' (Streams GPS coordinates)
Arrive at Customer Delivery Address
Delivery Outcome?
Successful Delivery
Collect COD Cash / Verify OTP
Customer Absent / Rejected
Record Failure Reason Code
Update Status to 'DELIVERED' (Emits realtime event to customer & merchant)
Update Status to 'RTO_INITIATED' (Return parcel to merchant warehouse)
Handover Collected Cash to Dispatcher